

## Formula SAE Task Project 1 – Datalogging Power Meter

**Subsystem(s):** Data Acquisition

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**Goal:**

Research and design a PCB that allows the collection of DC current and/or power levels from the battery and rectifier.

**Task Background:**

Current and voltage through the battery and rectifier are vital measurements that allow us to know the exact power consumption of the vehicle. While voltage is currently measured by a parallel circuit through the 12-14.3V battery, current usually must be measured either series to the circuit, or through a Hall Effect sensor. In our platform, the vehicle normally draws around 80-120A on startup, with a seized starter drawing upwards of 160A. During idle, the car draws around 16-22A, although this data fluctuates based on the duty cycles of powertrain parts. By creating an ammeter or power meter, we will be able to measure the exact power consumption and delta of the system.



*Figure 1: Ammeter Shunt*

**Task Description:**

The task is to design a datalogging PCB circuit that allows the collection of automotive current and/or power levels from the battery. It does not matter if you use an existing online circuit design, if you can find one. Additionally, you may use a microcontroller dev board (e.g. ESP-32) for the datalogging portion. Size constraints will be your greatest constraint.

Don't worry about temperature, just make sure there are no crazy hotspots on the board.

**Reminder:**

Remember to reach out to me ASAP for any additional resources or help that you may need. I have the CAD models, additional information, and possible measurements that are essential to this project. I can help clarify more about the fundamentals, or about what needs to be done by when. Don't like this project at all? Let me know and I'll switch you to a different project. It's not a problem – some don't like doing wiring harness work, many don't like coding, and others don't like working with control systems.